

# GPT Architecture and Evolution

## 1 Overview of GPT Architecture

GPT (Generative Pre-trained Transformer) models are based on the Transformer decoder architecture. The model is trained using a causal (unidirectional) language modeling objective, meaning each token is predicted based only on previous tokens.

**Key components:**

- **Token and positional embeddings:** Convert input tokens into vectors with position information
- **Stacked transformer decoder blocks:** Each includes self-attention, feed-forward layers, and residual connections
- **Masked self-attention:** Prevents attending to future tokens during training
- **Final linear layer + softmax:** Produces a probability distribution over the vocabulary

GPT models are autoregressive — they generate one token at a time from left to right.

## 2 GPT Model Evolution

### GPT-1

- Introduced in 2018 by OpenAI
- 117M parameters
- Trained on BooksCorpus

### GPT-2

- Released in 2019
- 124M to 1.5B parameters
- Trained on WebText

### GPT-3

- Released in 2020
- 175B parameters
- Introduced few-shot prompting

### GPT-3.5

- Released in 2022
- Used in early ChatGPT versions

## GPT-4

- Released in 2023
- Supports multimodal inputs
- Context windows up to 32K tokens

## 3 What Is ChatGPT?

ChatGPT is a conversational interface built on top of GPT models (3.5 and 4). It is designed for interactive, context-aware dialogue and instruction-following using a combination of pretraining and RLHF (Reinforcement Learning from Human Feedback).

## 4 Transformer Decoder Block: Internal Components

- **Masked multi-head self-attention:** Computes attention weights over previous tokens only; uses causal masking.
- **LayerNorm + residual connections:** Improves gradient flow and model stability.
- **Feed-forward network (MLP):** Applies nonlinear transformation to each position.
- **Output LayerNorm:** Final normalization before token logits are computed.

## 5 GPT vs BERT Comparison

| Feature       | GPT            | BERT           |
|---------------|----------------|----------------|
| Direction     | Unidirectional | Bidirectional  |
| Architecture  | Decoder-only   | Encoder-only   |
| Training Goal | Next token     | Masked token   |
| Use Case      | Generation     | Classification |

## 6 GPT Version Comparison Table

| Model   | Year | Parameters | Context Window | Key Features                  |
|---------|------|------------|----------------|-------------------------------|
| GPT-1   | 2018 | 117M       | ~512 tokens    | BooksCorpus, pretraining only |
| GPT-2   | 2019 | 1.5B       | 1024 tokens    | WebText, zero-shot            |
| GPT-3   | 2020 | 175B       | 2048 tokens    | Few-shot prompting, Codex     |
| GPT-3.5 | 2022 | ~175B      | 4096 tokens    | Dialogue tuning               |
| GPT-4   | 2023 | ?          | 8K-32K tokens  | Multimodal, strong reasoning  |

## 7 Training and Inference Workflow

### Training Phase

- Next-token prediction using cross-entropy loss
- Optimizers like Adam or AdamW
- Large-scale training data (e.g., Common Crawl, Wikipedia, books)

## Inference Phase

- Tokenize input
- Generate hidden states through decoder layers
- Output probability distribution over next tokens
- Apply decoding strategies: greedy, sampling, beam search

## 8 Limitations of GPT Models

- May hallucinate or invent facts
- Susceptible to bias in training data
- Limited by context window size
- Lacks real-time external grounding
- No long-term memory

## 9 Glossary of Key Terms

- **Token:** Smallest unit of text the model processes
- **Embedding:** Numerical vector representation of a token
- **Self-attention:** Mechanism to relate a token to all others
- **Logit:** Raw model output before softmax
- **Top-k sampling:** Sample from top k tokens
- **Beam search:** Keep top sequences across decoding paths
- **Temperature:** Controls randomness of generation
- **RLHF:** Human feedback to tune model alignment
- **Zero-shot:** No examples provided
- **Few-shot:** Prompt includes a few examples